

Multimodal Fashion & Context Retrieval

Approaches, Chosen Architecture, and Future Work

Glance ML Internship Assignment — Submission Write-up

1. Problem Summary

The task is to build a search engine that retrieves fashion images from natural-language descriptions that combine three things at once: what someone is wearing (garment type, colour, pattern), where they are (office, street, park, home), and the overall vibe of the outfit (formal, casual, professional). The assignment hint is explicit about the real challenge: a vanilla CLIP baseline handles single attributes reasonably well but is known to struggle with compositionality — binding the right colour to the right garment when a query names more than one — and with fine-grained fashion vocabulary that general-purpose pretraining under-represents.

This document covers, in order: the approaches considered and their tradeoffs, the architecture chosen and why it addresses CLIP's specific weaknesses, how it handles the assignment's five evaluation queries, and directions for future work (adding real locations/weather, and improving precision further).

2. Approaches Considered

Five candidate approaches were compared, ranging from a zero-effort baseline to a fully symbolic system, with the chosen hybrid sitting deliberately in between.

Approach	How it works	Strengths	Weaknesses	When to use
A. Vanilla CLIP (zero-shot)	Encode image and text query into one shared embedding space with an off-the-shelf pretrained CLIP; nearest-neighbour search on cosine similarity.	Zero setup cost, genuinely zero-shot, works on any vocabulary, fast to prototype.	Global pooled embedding entangles attributes ("red shirt, blue pants" and "blue shirt, red pants" land close together); weak on fine-grained fashion attributes and fabric/pattern detail.	Quick baseline, small datasets, non-compositional queries.
B. Fine-tuned CLIP (domain adaptation)	Continue contrastive training on fashion image-caption / image-attribute pairs (e.g. Fashionpedia captions) to pull the embedding space toward fashion vocabulary.	Better recall on fashion-specific terms (garment names, fabric, style); still a single fast embedding.	Still a single global vector per image — compositional binding problem is reduced, not solved; needs labeled fashion data and GPU training time; risks catastrophic forgetting of general vocabulary.	Medium-effort upgrade when the domain vocabulary is narrow and compute is available.
C. Structured / symbolic (detector + attributes only)	Run a garment detector (category + bounding box) and attribute classifiers (colour, pattern) over every image; store as structured fields; match queries by parsing them into the same schema and doing symbolic/SQL-style filtering.	Excellent precision on compositional queries; fully interpretable; cheap to query (no embedding math needed for known attributes).	Closed vocabulary — cannot handle attributes or phrasing outside the fixed taxonomy; loses "vibe"/style queries ("casual weekend outfit") that aren't literal attributes; no zero-shot generalization.	High-precision filtering layer, never a good standalone solution for open-ended natural language.

Approach	How it works	Strengths	Weaknesses	When to use
D. Hybrid: region-aware embeddings + retrieve-then-rerank (CHOSEN)	Detector crops each garment; each crop gets its own CLIP embedding, plus a separate scene embedding and a global embedding. Query is parsed into per-garment + scene + style slots. Stage 1 (ANN + metadata filter) gets recall; Stage 2 reranks the shortlist by matching each query slot to the correct image region.	Solves attribute-binding directly by keeping colour/garment pairs spatially separate; keeps CLIP's open-vocabulary zero-shot strength for style/vibe words; reranking is cheap because it only runs on a small shortlist, so it scales.	More moving parts to build and maintain than A or B; detector quality becomes a new failure mode; needs a lightweight query parser.	Best overall fit for this assignment's stated goal: better than vanilla CLIP on compositionality, while staying practical to build and scale.
E. Full cross-attention VLM reranker (BLIP-2 / LLaVA style)	A vision-language model jointly attends over image and text and outputs a relevance/match score directly, rather than comparing two independently-pooled embeddings.	Potentially the strongest compositional and relational reasoning (e.g. “tucked into”, “over a hoodie”) because it never throws away cross-modal attention.	Cannot be indexed — needs a forward pass per (query, image) pair, so it cannot search a large corpus directly; heavy latency/cost at scale.	As an optional Stage 3 reranker on the top ~10–20 candidates only, if budget allows — not as the primary index.

The core tension across all five is the same one CLIP research keeps running into: pooled, single-vector embeddings are cheap and generalize well (zero-shot), but they average away the spatial binding between an attribute and the object it belongs to. Symbolic/structured systems fix binding perfectly but only for attributes they were built to recognize. The chosen approach (D) keeps both: CLIP's open vocabulary for style and unseen combinations, and localized, per-garment matching for the compositional cases the assignment specifically calls out.

3. Chosen Approach: Region-Aware Compositional Retrieval with Retrieve-then-Rerank

3.1 Why this, specifically

Vanilla CLIP represents a whole image as one pooled vector. When a query names two garments with two colours, that single vector has no mechanism to keep “red” bound to “tie” and “white” bound to “shirt” rather than the reverse — this is the same bag-of-attributes weakness documented in compositional vision-language benchmarks (e.g. ARO, SugarCrepe-style evaluations) and is exactly the failure mode the assignment hint warns about. Fine-tuning CLIP on fashion captions (Approach B) improves fashion vocabulary coverage but does not change the fact that it is still one vector per image — it narrows the problem without solving it.

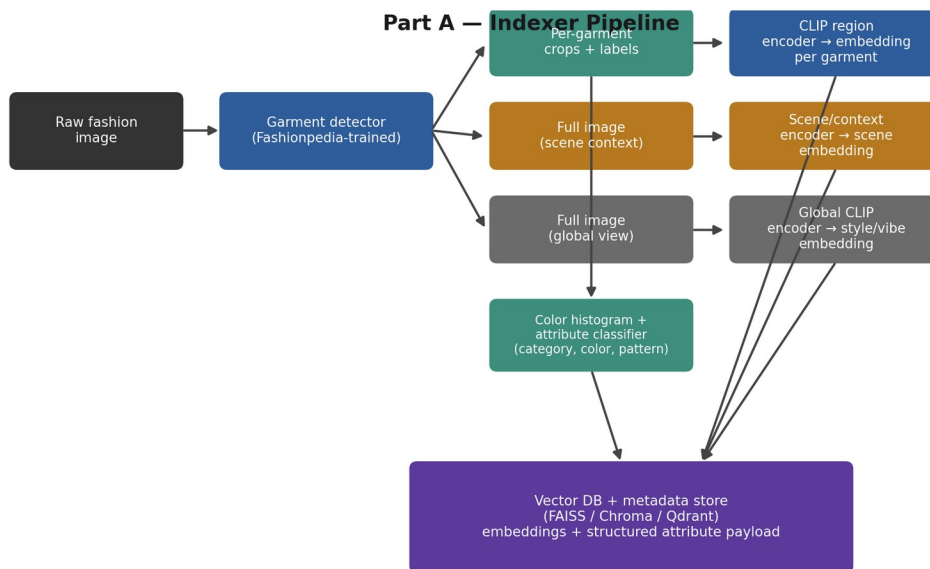
The fix chosen here is architectural, not just a better-trained encoder: detect and crop each garment first, embed each crop separately, and only then match a multi-attribute query slot-by-slot against the right region. A separate scene/context branch handles “where,” and the original global CLIP embedding is kept as a fallback / style channel so the system does not lose zero-shot ability on descriptions and vocabulary it has never explicitly seen labeled.

3.2 Indexer architecture (Part A)

Each image is processed once, offline, into a small set of stored vectors plus structured metadata:

- A garment detector (fine-tuned on a fashion-labeled set such as Fashionpedia, which already provides 46 categories and ~92 fine-grained attributes) finds each garment's bounding box and category.

- Each garment crop is passed through a CLIP image encoder independently, producing one embedding per garment — this is what keeps colour bound to the correct object.
- The colour/pattern for each crop is extracted with a lightweight attribute classifier (or a simple dominant-colour histogram over the masked region as a cheap fallback), stored as structured metadata alongside the embedding.
- The full image is also passed through a scene/context branch (a CLIP embedding computed over the whole frame, optionally combined with a scene classifier such as a Places365-style model) to capture “office / park / street / home.”
- A global CLIP embedding of the full image is kept as well, specifically to preserve open-vocabulary style/vibe matching (“casual,” “professional”) that isn’t a fixed attribute.
- All vectors plus the structured attribute payload are written to a vector database that supports combined vector + metadata (payload) search — per the assignment’s own guidance, the simplest adequate option is used rather than a custom index: Chroma or Qdrant both support this out of the box with a few lines of client code, so no custom ANN implementation is needed.

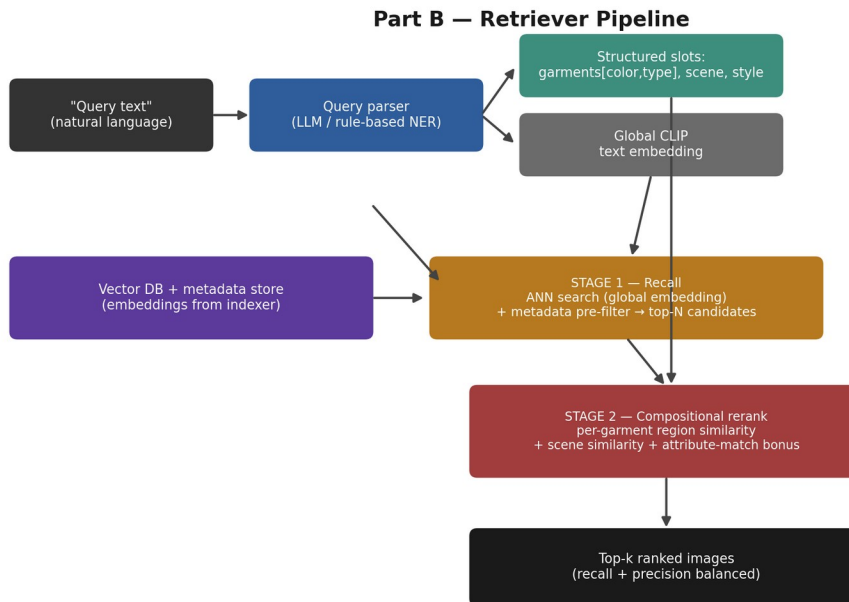


3.3 Retriever architecture (Part B)

At query time, natural language is turned into a shortlist and then re-scored in two stages so that the expensive, compositional part of the work only ever runs on a small candidate set:

- A lightweight query parser (either a small prompt to an LLM, or a rule-based colour/garment/location extractor as a cheaper fallback) turns the raw query into structured slots: a list of {garment type, colour} pairs, a scene/location phrase, and a residual style phrase for anything that doesn’t fit a slot.
- The same query is also embedded directly with the CLIP text encoder, so the open-vocabulary path is never lost even if the parser misses something.
- Stage 1 (recall): an approximate nearest-neighbour search against the global embeddings, optionally pre-filtered by any hard metadata constraints the parser was confident about, returns a shortlist (e.g. top 200) — this is the only stage that needs to scale to the full corpus.
- Stage 2 (precision / rerank): for just that shortlist, each parsed garment slot is matched against the correct garment-region embeddings of each candidate image, the scene slot is matched against the scene embedding, and an attribute-match bonus is added when the structured metadata agrees exactly (e.g.

classifier says “yellow raincoat” and the query says “yellow raincoat”). These are combined into one weighted score and the top-k is returned.



3.4 Walking through the assignment's evaluation queries

The five queries in the assignment brief exercise different parts of the pipeline; going through them explicitly is a useful check on what the design does and does not solve.

Evaluation query	Primary signal path	Why it works
“A person in a bright yellow raincoat.”	Detector finds outerwear region → attribute classifier confirms colour = yellow, category = raincoat → region embedding + attribute match score high.	Single garment, single colour — the easy case for both baseline CLIP and the hybrid system; mostly a sanity check.
“Professional business attire inside a modern office.”	Scene branch matches “office”; garment-category attributes (blazer / button-down) matched against “formal” cluster; global CLIP embedding contributes the “professional” style/vibe signal.	Needs both context (place) and style inference together — exactly the two-branch (scene + style) design this hybrid keeps separate then fuses.
“Someone wearing a blue shirt sitting on a park bench.”	Garment slot (blue, shirt) matched at region level; scene branch matches “park”; the “sitting on a bench” detail is handled at the scene/global level (park-bench co-occurrence, object cues) rather than as a garment attribute.	Shows a known limitation: pose/action grounding is weaker than garment-colour grounding — flagged explicitly in Future Work rather than overclaimed.
“Casual weekend outfit for a city walk.”	No literal colour/garment named — relies mostly on the global CLIP style embedding (casual vs. formal cluster) plus scene branch (urban street).	Pure style-inference query with no explicit attributes — the case where keeping a global, zero-shot CLIP embedding in the mix (rather than going fully symbolic) actually pays off.
“A red tie and a white shirt in a formal setting.”	Two garment slots parsed separately (red / tie, white / shirt), each matched to its own detected region; scene/style branch confirms “formal.”	The textbook compositional-binding test from the assignment hint — this is exactly the failure mode a single pooled CLIP vector is prone to, and exactly what per-region matching is designed to fix.

3.5 Vector storage and scalability

Per the assignment's own steer — spend engineering effort on the ML logic, not on reinventing indexing — the plan is to use an existing embedded/managed vector database (Chroma for the simplest local setup, or Qdrant if payload filtering and horizontal scaling matter more) rather than a hand-rolled ANN index. This keeps Part A's deliverable focused on getting the embeddings and attributes right rather than on index engineering.

At 1M+ images, three things keep this practical:

- Stage 1 always searches only the compact global-embedding index (one vector per image), which is exactly what FAISS/HNSW-backed vector databases are built to scale to millions of entries with sub-100ms latency.
- Stage 2's more expensive per-garment matching only ever runs on the Stage-1 shortlist (tens to a few hundred candidates), so its cost is constant with respect to corpus size, not linear in it.
- Metadata payload filters (category, colour, scene tag) can be pushed down into the vector database's own filtered search, shrinking the candidate set before any embedding math runs at all — both Chroma and Qdrant support this natively.

4. Codebase

GitHub repository link: [to be added once the implementation is pushed — this write-up was scoped, per request, to the approach and architecture rather than the runnable implementation].

For reference, the intended repository layout keeps indexing and retrieval as clearly separate modules, as the assignment asks:

- `indexer/detect.py` — garment detection + cropping
- `indexer/embed.py` — region, scene, and global CLIP embedding extraction
- `indexer/attributes.py` — colour/pattern attribute extraction
- `indexer/build_index.py` — writes embeddings + metadata into the vector database
- `retriever/parse_query.py` — natural-language query → structured slots
- `retriever/search.py` — Stage 1 ANN recall + Stage 2 compositional rerank
- `retriever/evaluate.py` — runs the five evaluation queries (and any others) and reports top-k results
- `common/` — shared config and the vector-database client wrapper

5. Future Work

5.1 Extending to real locations (cities, places) and weather

The current scene branch only recognizes generic categories (office, park, street, home) that are visually learnable from the image itself. Two extensions would take this further:

- Named places (a specific city or landmark) generally cannot be inferred from clothing/background alone with any reliability — this needs either photo metadata (EXIF GPS, or a location tag supplied at ingestion time) stored as a searchable payload field, or a dedicated landmark-recognition model for the subset of images that do contain identifiable landmarks.
- Weather can be partly inferred visually (a dedicated weather/scene classifier trained on categories like rainy street, snowy park, sunny beach), and partly inferred indirectly from clothing correlation — a raincoat plus

umbrella is itself weak evidence of rain — so the two signals should be fused rather than treated as independent.

- Both extensions are additive: they add new metadata fields and, optionally, a new embedding branch, without requiring any change to the Stage 1 / Stage 2 retrieval logic already in place.

5.2 Improving precision further

- Fine-tune the garment detector on pixel-level segmentation masks (Fashionpedia provides these) rather than bounding boxes, so colour/pattern extraction doesn't bleed in background pixels at the crop edges.
- If any CLIP fine-tuning is done, construct hard negative pairs deliberately — the same two garments with colours swapped — so the model is explicitly trained on the exact compositional failure mode the assignment calls out, rather than hoping it emerges from generic contrastive pretraining.
- Add a lightweight learned reranker (e.g. a small cross-encoder or gradient-boosted model over the Stage-2 similarity features) once real query-click or relevance-judgment data is available, rather than hand-weighting the Stage 2 score components indefinitely.
- Tune the vector index itself for the recall/precision/latency tradeoff at scale (e.g. HNSW ef_search, or IVF-PQ nlist/nprobe) once the corpus size and latency budget are known.
- Cache embeddings for frequent or repeated query phrases, and consider an optional Stage 3 cross-attention VLM reranker (Approach E) on just the final top 10–20 candidates for queries that involve relational phrasing (“tucked into,” “layered over”) that pure similarity scoring under-handles.

6. Summary Against the Assignment's Evaluation Criteria

- **Modular code:** indexing and retrieval are separate modules with no shared mutable state; the vector database is the only interface between them.
- **Scalability to ~1M images:** only Stage 1 touches the full corpus, and it uses an off-the-shelf ANN-backed vector database rather than a custom index; Stage 2's cost is bounded by the shortlist size, not corpus size.
- **Zero-shot capability:** the global CLIP embedding and text encoder are kept in the loop specifically so unseen attribute combinations and style phrasing still get scored, rather than falling back to a closed symbolic taxonomy.
- **Better than vanilla CLIP on compositionality:** addressed structurally (per-garment regions matched to per-garment query slots) rather than by hoping a bigger or fine-tuned encoder learns binding implicitly.